Proofs and Usages of the Cantor-Bernstein-Schröder Theorem

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September 2026

Abstract

This article aims to include a gentle revision for the concept that treats function as a specific kind of relations, the latter of which are but sets with ordered pairs. Then, this article also gives a proof of one of the most famous consequences in Elementary Set Theory, the Cantor-Bernstein-Schröder Theorem. Finally, some useful yet immediate corollaries are given. Diagrams are used extensively as a means of conveying informtion more concisely.

1 Introduction

We are extremely accustomed to the basic properties of real numbers, namely

$$\forall \alpha, \beta \in \mathbb{R}, \alpha \leq \beta \wedge \beta \leq \alpha \implies \alpha = \beta.$$

This is but a more 'mathy' way of saying that *if there's a pair of real numbers that both of them satisfy the statement 'this number is less than or equal to the other', then they must be equal.* This is so intuitive that some real analysis textbooks would prefer omitting it to devoting time to it by proving it from some more fundamental fields axioms. We would like to omit the proof because this article is supposed to dedicate to Set Theory.

Remark 1. For those that are very curious about properties of real numbers or would like to build intuition about the features of strict inequality we would like to employ later on, or are just wondering what Real Analysis is like, the author would like to present a disproof of a slightly altered statement, so that we may skip the discussion about the impossibility of strict less-than in the following excursions (the following proof is, of course, not extremely rigorous, but it should be enough for the rest of the article).

We would like to prove that there does not exist real numbers α and β such that $\alpha < \beta$ and $\beta < \alpha$.

Proof. Indeed, one popular definition of the order relation on real numbers $a < b$, $a, b \in \mathbb{R}$ is that b can be rewritten as $b = a + \varepsilon$, for $\varepsilon > 0$.

Then write $\beta = \alpha + \varepsilon_1$ given that $\alpha < \beta$, and write $\alpha = \beta + \varepsilon_2$ given that $\beta < \alpha$. Both errors are strictly positive. Then utilise the equality with subject of β into the latter equation, and it yields $\alpha = \alpha + \varepsilon_1 + \varepsilon_2$. Technically one should prove arduously many field laws to guarantee that cancellation works, but the author would like to omit so, and it still yields $0 = \varepsilon_1 + \varepsilon_2$. Again, it is not very hard to tell that this is an impossible scenario: the sum of two strictly positive integers gives a nonpositive zero. □

Remark 2. This is somehow the author's own 'advertisement' for Real Analysis, but a true real analyst deals with series formally (usually comprising many infinitely small quantities) as well as differentiation and integration instead of individual chunks of real numbers. But real analysts do manipulate inequalities very frequently; people often banter that it is the 'pedantic version of calculus'.

2 A Closer Look at Sets

Anyway, let us rectify our digression and step back into Set Theory. There is a strikingly similar argument that also originates from Set Theory to the theorem we would like to explore below. It writes:

Theorem 1 (Antisymmetric property of set inclusion)**.** *Do not be intimidated by its name. It basically just states that*

$$A \subseteq B \wedge B \subseteq A \iff A = B.$$

Proof. Its proof is also supposed to be very straightforward.

($\Longrightarrow$): Given that $A \subseteq B$, every single element in A is in B; given that $B \subseteq A$, every single element in B is in A. Then they both contain the exactly same element.

($\Longleftarrow$): Given that $A = B$, we may as well '*regurgitate*' the exact corollary immediately from the definition ditto. □

This would also be very obvious when two sets, irrespective of cardinality, are given:

Example 1. We see that $\{1, 2, 3\}$ is an improper subset of $\{3, 2, 1\}$ (recall: order in ordinary sets does not matter), and vice versa. So, they are equal.

Example 2. If we adopt the definition where 0 is **not** included (which is frequently used in Real Analysis but not that frequent in Algebra) in $\mathbb{N}$, then because $\mathbb{Z}^+ \subseteq \mathbb{N}$ and $\mathbb{N} \subseteq \mathbb{Z}^+$, $\mathbb{Z}^+ = \mathbb{N}$.

The two sets here have a countably infinite cardinality, yet they are the same thanks to the mutual inclusion expressed via the means of being mutual subsets.

However, it is the strict premise that mutual inclusion must be attained that makes the proof and examples look very trivial to examine the two sets' equality. By considering purely the *cardinality*, the same observation, albeit still true, would be very interesting and nontrivial to handle.

Nonetheless, for finite sets, one can still intuitively come up with such a phenomenon:

Theorem 2. *If two finite sets A and B have the cardinality $|A|$ and $|B|$, and that $|A| \leq |B|$ as well as $|B| \leq |A|$, then $|A| = |B|$.*

This is, in fact, already the finite version of Cantor-Bernstein-Schröder Theorem; therefore, what we are essentially doing here is already partially proving the theorem we are going to discuss.

Proof. By the theorem we have just gone through on real numbers, it suffices to conclude that $|A| = |B|$ given the premises. Alternatively, one may just treat the proof with rigour by adding the line of letting some integer r be the cardinality of one of the two sets, and use r as the target of comparison instead of cardinalities themselves (after all, all integers are real numbers). □

Example 3. Using $\{1, 3, 5, 7, 9\}$ and $\{2, 4, 6, 8, 10\}$, for instance, we can immediately tell that they share a cardinality of five, although they do not share identical elements. However, it suffices to conclude that their cardinality is the same.

However, according to a popular Russian literature [1, p. 23], one should should not treat cardinalities of infinite sets (be they countable or not) as standalone, tangible numbers, but instead as relative concepts that could only be compared against. Unfortunately (but it is what brings it more 'fun'), set theorists would mainly like to deal with infinite sets. So, we need a even more robust 'backbone' to back our excursion; namely, a more holistic definition of functions as relations.

3 A Closer Look at Functions

In elementary mathematics one is often taught that *machines* are good analogies to rudimentary mathematical functions: you take in some value that is 'satisfied' or 'accepted' by the function, and the latter always outputs exactly one single value. However, this is not very formal. So we must proceed with a new definition, which will be a bit overwhelming but is extremely important in proving our theorem:

Definition 1. For any set A and B, a function (f here) sending (formally *mapping*) elements in A to elements in B is denoted by $f : A \mapsto B$. Here, A is called the *domain* and B the *codomain*.

It is a relation from A to B that for every $x \in A$, there exists uniquely a $y \in B$ such that $x \sim y$.

Here, the relation is just a set with 'many' ordered pairs; a special property of this relation is that the slot of x in every single pair is unique:

$$f = \{\langle x_1, y_1\rangle, \langle x_2, y_2\rangle, \cdots\}.$$

The Cantor-Bernstein-Schröder Theorem is mainly concerned with the bijectivity between two sets, so we shall define this kind of special properties of functions (with f as the example ditto):

Definition 2. f is *injective* if

$$\forall x_1, x_2 \in A, f(x_1) = f(x_2) \implies x_1 = x_2.$$

In other words, every single value is mapped to a unique value on the other side.

Definition 3. f is *surjective* if

$$\forall y \in B, \exists x \in A\big(f(x) = y\big).$$

In other words, every single y in the destination set B can be reached by some value in A.

Definition 4. f is *bijective* if it is both injective and surjective.

Then recall the definition of the inverse of a function, which in elementary mathematics we are used to 'swapping' the two variables in an expression to find (again, continuing the same f):

Definition 5. If there exists a function $g : B \mapsto A$ that it satisfies:

- $\forall x \in A, (g \circ f)(x) = x$,
- $\forall y \in B, (f \circ g)(y) = y$,

Then g is the inverse of f. Write g as f^{-1}; where the operator $\circ$ is the symbol denoting the composition of functions: $(g \circ f)(x) = g(f(x))$.

Before (finally) proving our theorem, let us revisit the concept of image and preimages, which are essentially ways to formalise some special behaviours of a well-defined function if only a 'portion' of elements are extracted from its domain and codomain, called the image and the preimage of the function respectively. Academically, for sets $A', A'' \subseteq A$ and $B', B'' \subseteq B$, with the same definition of the function f ditto,

Definition 6. $\{f(x) \in B : x \in A'\}$ (written as $f(A')$) is called the *image* of A', and $\{x \in A : f(x) \in B'\}$ (written as $f^{-1}(B)$) is called the *preimage* of B'. This notation is still used even a well-defined, proper inverse of f does not exist (notice that B is a set closely related to the entire codomain of the function, and is not a single value to be mapped to).

Then, it follows that,

Theorem 3. *Continuing the definition, the following expressions hold:*

- $f(A' \cup A'') = f(A') \cup f(A'')$
- $f(A' \cap A'') = f(A') \cup f(A'')$
- $f^{-1}(B' \cup B'') = f^{-1}(B') \cup f^{-1}(B'')$
- $f^{-1}(B' \cap B'') = f^{-1}(B') \cup f^{-1}(B'')$
- $f(f^{-1}(B')) \subseteq B'$
- $f^{-1}(f(A')) \supseteq A'$

Sketching the proof. We only need the last two to prove the Cantor-Bernstein-Schröder Theorem, so we would omit the proof of the first four. Readers may treat them as exercise at your own will, but the author does not intentionally *leave them as exercise to the readers.*

It may be very counterintuitive to see that the composition of the inverse of the function and the function itself, in both orders, does not (always) yield a strict equality. This can be ascribed to our instinct of seeing the sets as individual values cherrypicked from the domain or codomain, but what we are dealing with now is the whole subset, which may even be infinite.

So to ditch this misconception, the best way to think is to utilise the definition of surjectivity and injectivity when handling those sets. See below. □

Proof. ($f(f^{-1}(B') \subseteq B'$): Let y be an arbitrary element in the image of the preimage such that $y \in f(f^{-1}(B'))$. Then by definition of this whole chunk of function, there exists some element $x \in f^{-1}(B')$ such that $f(x) = y$. Since $x \in f^{-1}(B')$, it follows from the definition of preimage that its output under f must lie in B' (i.e., $f(x) \in B'$). Because $y = f(x)$ and $f(x) \in B'$, then $y \in B'$. But y is arbitrarily picked, so for any element in B' it holds.

Similarly, let x be an arbitrary element in the initial set such that $x \in A'$. To show that x belongs to the preimage $f^{-1}f(A')$), showing its image $f(x)$ belonging to the set $f(A')$ is sufficient. By definition (of the image), $f(x) \in f(A')$. Then, by definition (of the preimage), the element x satisfies that $x \in f^{-1}(f(A'))$. Since every $x \in A'$ is also in $f^{-1}(f(A'))$, we have $f^{-1}(f(A')) \supseteq A'$. □

The diagram below visualises the concept.

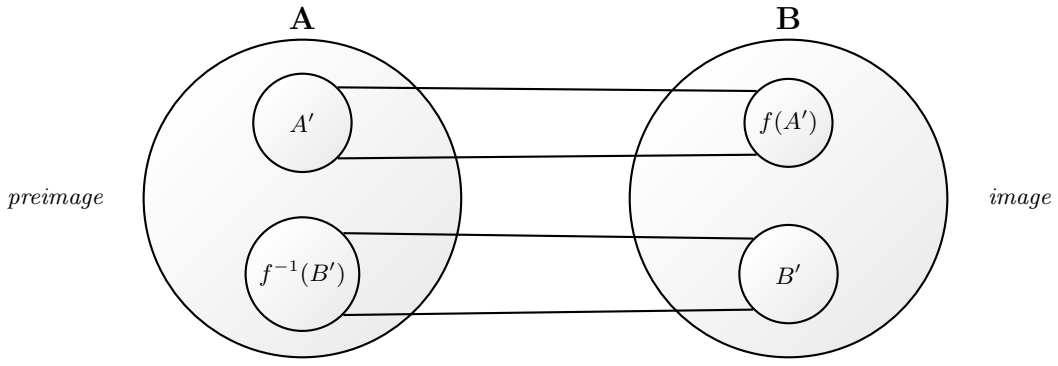


Figure 1: The relationship between preimage and image sets

Theorem 4. *A key realisation, perhaps with the aid of the figure, is that a strict equality (instead of relations of supersets or subsets) holds if and only if f is bijective. Specifically, the $f(f^{-1}(B')) = B'$ (holds for all subsets $B' \subseteq B$) if and only if f is surjective. Similarly, $f^{-1}(f(A')) = A'$ for all $A' \subseteq A$ if and only if f is injective. This is why one would soon see sometimes we use the equal sign and superset/ subset signs otherwise in the following principal proof of the Cantor-Bernstein-Schröder Theorem.*

Proof. We only need to show $B' \subseteq f(f^{-1}(B'))$, then we can apply the above lemma we have already discussed. Let $y \in B'$. Since f is surjective, an $x \in A$ must exist such that $f(x) = y$. Hence $f(x) \in B'$, meaning $x \in f^{-1}(B')$ by the definition of preimage. Thus, its own image $f(x)$ belongs to $f(f^{-1}(B'))$. Since $y = f(x)$, $y \in f(f^{-1}(B'))$, thus $B' \subseteq f(f^{-1}(B'))$.

Similarly, to show $f^{-1}(f(A')) \subseteq A'$ when f is injective, Let $x \in f^{-1}(f(A'))$. Then $f(x) \in f(A')$ straight from the definition. By definition of image, $f(x) \in f(A') \implies \exists x' \in A' : f(x) = f(x')$. Guaranteed by the injectivity, cancel the f to get $x = x'$. But $x' \in A'$, so $x \in A'$. □

If the above feels very alien to the readers, the author would like to present an example illustrating the concept.

Example 4. Let there be a function with finite domain $\{1,2,3,4\}$ and infinite range $\mathbb{R}\setminus\mathbb{Q}$. This is contrived to be so for the sake of simplicity, yet it should work regardless of the cardinalities of the domain or the range.

It maps with the following scheme: $1\mapsto e, 2\mapsto e, 3\mapsto\pi$ and $4\mapsto\pi$. Clearly the function is neither injective nor surjective, since both one and two are mapped to e, and values such as $\sqrt{2}$ are irrelevant in the mapping anywhere, yet they are valid members of the codomain.

Now take $A'=\{1,3\}$ and $B'=\{e,\pi,\sqrt{2}\}$. Then $f^{-1}(B')=\{1,2,3,4\}$. Compute $f(f^{-1}(B'))=\{e,e,\pi,\pi\}=\{e,\pi\}$. Note that $\sqrt{2}$ is omitted because this constant is irrelevant to the definition. Similarly, for $f(A')=\{e,\pi\}$, and its composite $f^{-1}(f(A'))$ yields $\{1,2,3,4\}=A$. This is because there is *more than one* 'input' mapping to one single 'output' (it is thus not injective), and because it includes every single possible value pointing to those specific *outputs*, it ineluctably enlarges the resulting set.

Although tricky, a thorough and solid understanding of the above peculiar properties is required to understand the upper half of the imminent proof below.

4 Scrutinising the Infinite Sets

Theorem 5 (Cantor-Bernstein-Schröder)**.** *For any two sets A and B, irrespective of cardinality, if an injection from A to B can be constructed, and so is an injection from B to A, then there exists a well-defined bijection between A and B.*

Proof. Let $\sigma: A\mapsto B$ and $\tau: B\mapsto A$, where σ and τ are injections. Then one claims that $|A|\leq|B|$. Indeed, one guarantees that there exists a σ to ensure $x_1=x_2 \iff \sigma(x_1)=\sigma(x_2)$. It can be given that with the aid of the description in the theorem, one can readily derive that $|B|\leq|A|$ too.

Then, let $A_0=A$ and $B_0=B$ where $\forall n\in\mathbb{N}$, $A_n=(\tau\circ\sigma)(A_{n-1})$ and $B_n=(\sigma\circ\tau)(B_{n-1})$.

It immediately follows that $A_0\supseteq A_1\supseteq A_2\supseteq\cdots$ and $B_0\supseteq B_1\supseteq B_2\supseteq\cdots$, because for any set S (here comes a dummy variable irrelevant to the discussion hitherto), $f^{-1}(f(S))\supseteq S$ (indeed, we have just proved so and the readers are reminded to have a 'thorough understanding'). Specifically, $\sigma(A_0)\subseteq B_{n-1}$, $\tau(B_{n-1})\subseteq A_{n-1}$. Then,

$$\begin{aligned}
\implies \sigma(A_n)&=(\sigma\circ\tau\circ\sigma)(A_{n-1})\subseteq(\sigma\circ\tau)(B_{n-1})=B_n,\\
\tau(B_n)&=(\tau\circ\sigma\circ\tau)(B_{n-1})\subseteq(\tau\circ\sigma)(A_{n-1})=A_n,\\
\therefore\forall n\in\mathbb{N},\quad &\sigma(A_n)\subseteq B_n,\quad \tau(B_n)\subseteq A_n,\ \text{by induction:}
\end{aligned}$$

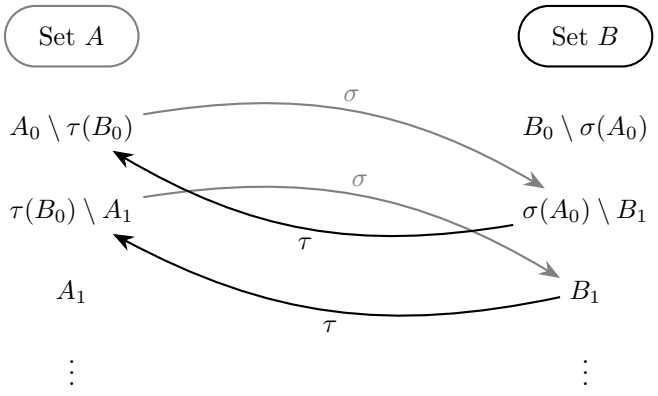


Figure 2: The relation demonstrated behind this theorem, the *Cantor-Bernstein-Schröder Theorem*

Then, let

$$A_\infty=\bigcap_{i\in\mathbb{N}}A_i,$$
$$B_\infty=\bigcap_{i\in\mathbb{N}}B_i.$$

If there exists $x\in A_\infty$ then $\forall n\in\mathbb{N}\big(\sigma(x)\in B_n\big)\implies\sigma(x)\in B_\infty$. It is similar for B_∞, and hence one can derive that

$$\sigma(A_\infty)\subseteq B_\infty,$$
$$\tau(B_\infty)\subseteq A_\infty.$$

Since $\sigma: A\mapsto B$ is an injection,

$$\forall n\in\mathbb{N}\begin{cases}\sigma_n: A_n\mapsto B_n & \text{is an injection,}\\ \sigma_n: A_n\mapsto\sigma(A_n) & \text{is a bijection,}\\ \sigma_n: A_n\setminus\tau(B_n)\mapsto\sigma(A_n)\mapsto B_{n+1} & \text{is a bijection.}\end{cases}$$

Similarly,

$$\forall n \in \mathbb{N} \begin{cases} \tau_n : B_n \mapsto A_n & \text{is an injection,} \\ \tau_n : B_n \mapsto \tau(B_n) & \text{is a bijection,} \\ \tau_n^{-1} : \tau(B_n) \mapsto B_n & \text{is a bijection,} \\ \tau_n^{-1} : \tau(B_n) \setminus A_{n+1} \mapsto B_n \setminus \tau(A_n) & \text{is a bijection.} \end{cases}$$

Then, we now define

$$v : A \mapsto B \begin{cases} \sigma(x) & x \in A_n \setminus \tau(B_n) \\ \tau^{-1}(x) & x \in \tau(B_n) \setminus A_{n+1} \\ \sigma(x) & x \in A_\infty \end{cases}$$

Intuitively, $u : A \setminus A_\infty \mapsto B \setminus B_\infty$ is a bijection (for the definition of λ, see the discussion near the end of this proof):

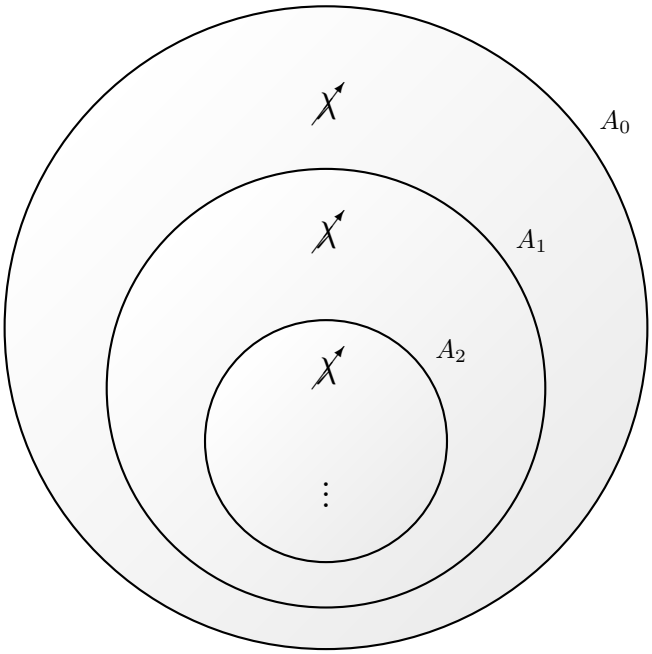


Figure 3: The rationale behind $\lambda \longrightarrow \kappa$

If the domain and the codomain are $A_\infty \mapsto B_\infty$ respectively, then $v = \sigma$, and so it is intuitive (right from the definition). Without loss of generality, assume then, for the sake of the proof, that $v : A_\infty \mapsto B_\infty$ is not surjective.

It is apparent that

$$\exists \kappa \in B_\infty, \quad \kappa \notin \sigma(A_\infty).$$

Thus,

$$\begin{aligned} &\implies \kappa \in B_\infty \implies \forall n \in \mathbb{Z}^+ \big(\kappa \in B_n = (\sigma \circ \tau)(B_{n-1}) \subseteq \sigma(A_{n-1})\big) \\ &\implies \forall n \in \mathbb{N}, \quad \kappa \in \sigma(A_n) \end{aligned}$$

Since σ is injective, $\forall n \in \mathbb{N}$ there exists only one $x_n \in A_n$ where $\sigma(x_n) = \kappa$. This implies that we can take a λ, where $\lambda = x_0 = x_1 = x_2 = \cdots$. Hence

$$\forall n \in \mathbb{N}, \quad \lambda \in A_n \implies \lambda \in A_\infty \implies \kappa = \sigma(\lambda) \in \sigma(A_\infty)$$

We have reached a contradiction that v must be bijective, and for this very bijection is the mapping we are looking. □

5 Usages of the Theorem

We are going to look at some uses of the Cantor-Bernstein-Schröder Theorem.

Example 5 (Equivalence of Intervals)**.** We want to prove that $(0, 1) \simeq [0, 1]$.

Proof. Without Cantor-Bernstein-Schröder Theorem, we are quite constrained because we are to find a perfect bijection to support the claim (so that we may prove straight from the definition). However, it is very trivial with the Cantor-Bernstein-Schröder Theorem. (Without the Theorem, we may even need to construct a piecewise function.)

We construct the identity as the first injection $\sigma(x) = x$, from $(0, 1)$ to $[0, 1]$. Then we also find the linear function $\tau(x) = \frac{x}{2} + \frac{1}{4}$ to restrict the domain and codomain from $[0, 1]$ and $[\frac{1}{4}, \frac{3}{4}] \in (0, 1)$. Note that the latter holds because injection carries transitivity (in an informal, intuition-based way of saying) and does not have to be exactly mutually equal to the two sets. Then according to the Theorem, they have equal cardinality. □

Example 6 (Finite dimension)**.** This shall be one of the most astonishing consequences. Intuitively, a unit square should possess 'more points' than the unit line, and a unit cube should possess 'more points' than the unit square. However, it turns out that they all bear the identical cardinality.

Mathematically, we would like to claim that $|\mathbb{R}| = |\mathbb{R}^2| = \cdots$.

Proof. We construct injection $\sigma(x) : \mathbb{R} \mapsto \mathbb{R}^2 = x \mapsto (x, 0)$. Then here comes another injection $\tau(x) : \mathbb{R}^2 \mapsto \mathbb{R}$ as follows: interleaving the decimal representation betwixt $0.a_1a_2a_3\ldots$ and $0.b_1b_2b_3\ldots$ as $0.a_1b_1a_2b_2a_3b_3\ldots$. We would like to omit the proof of handling infinitely repeating nines because we seek merely an injection, and we would also like to omit the proof by induction to promote the case from $\mathbb{R}^2$ to $\mathbb{R}^3, \mathbb{R}^4$, etc., because it is relatively trivial. □

An interesting example is disproving the following fallacious 'disproof' of the well-known Continuum Hypothesis.

Hypothesis (The Continuum Hypothesis)**.** *Let the cardinality of* $\mathbb{N}$ *be* $\aleph_0$*. Then the identity* $2^{\aleph_0} = \mathfrak{c} = |\mathbb{R}|$ *is well-established, and the Continuum Hypothesis argues that* $\mathfrak{c} = \aleph_1$*.*

The fallacy below is somewhat common amongst novices in Elementary Set Theory.

Fallacy ('Disproof' of the Continuum Hypothesis)**.** *Construct* $\mathbb{R} \setminus \mathbb{N}$*, then* $|\mathbb{N}| < |\mathbb{R} \setminus \mathbb{N}| < |\mathbb{R}|$*.*

Remark 3. The Continuum Hypothesis is destined to be forever standing as an unproven 'hypothesis', as both Cohen and Gödel had proved that the Continuum Hypothesis can both be accepted and rejected without causing much harm to Set Theory. Informally, their proofs can be summarised as follows: assume the acceptance or denial of the Continuum Hypothesis. No matter which path they had chosen, no contradiction would it lead to against any known axioms in Set Theory, at all. Hence this has philosophically been transformed to a matter of choice instead of a firm, absolute theorem.

It is also worth noting that this Gödel is the mathematician who had also proved that any mathematical system powerful to the extent of expressing basic arithmetic cannot be both complete and consistent. Combining the former and latter two 'ideal' scenarios gives that there does not exist any true statements unprovable, which is not the case. It is so interesting that it worths heated discussing, but doing so will lead us astray too outrageously.

Proof. Please do not be intimidated by the remark above; they are meant to be supplementary materials. The disproof specifically against the above fallacy is actually very trivial with the help of the Cantor-Bernstein-Schröder theorem.

We first define the first injection from $\mathbb{R} \setminus \mathbb{N}$ to $\mathbb{R}$. It turns out that the identity $\sigma(x) = x$ suffices.

Then we construct another injection strictly without any natural numbers, for instance,

$$\tau(x) = \frac{1}{\pi} \arctan x + \frac{1}{2}$$

from $\mathbb{R}$ to $(0, 1)$. Then according to the Cantor-Bernstein-Schröder Theorem, $|\mathbb{R} \setminus \mathbb{N}| = |\mathbb{R}|$. □

6 Conclusion

We have already seen the might of this single theorem guaranteeing equal cardinality between two sets, provided that two injections can be built. There is a so-called 'trick' (but it does not always work) of proving so using the Cantor-Bernstein-Schröder Theorem, namely, (we have already used it a few times) when encountering two sets with strikingly similar definitions, we may as well use the identity function $\sigma(x) = x$ as one of the two injections. Otherwise, to be optimistic, consider that the way via Cantor-Bernstein-Schröder Theorem is still better than finding a perfect bijection.

This article also reminds the readers that we should never treat infinite quantities identically as those finite figures. More fundamentally, bear in mind that all the infinities are concepts instead of real, concrete numbers. It is also this reason that explains why we need to be '*so rigorous*' that we insist in proving the complete set of Cantor-Bernstein-Schröder Theorem even the finite portion had been tackled.

References

[1] Shen and Vereshchagin. *Basic Set Theory.* The American Mathematical Society, 2002.